

# SECHINDRA VALLURY

 Odum School of Ecology, University of Georgia

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## Education

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- 2014-2019 School of Sustainability, Arizona State University, AZ.  
Ph.D., Sustainability. Advisors: Drs. Joshua K Abbott & John M Anderies.
- 2012-2014 Department of Natural Resources, TERI School of Advanced Studies, India.  
M.S., Environmental Studies and Resource Management, May 2014.
- 2006-2010 Department of Mechanical Engineering, Jawaharlal Nehru Technological University, India. B.S., Mechanical Engineering, May 2010.

## Professional Experience

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- 08/22- Assistant Professor of Environmental Policy, Odum School of Ecology  
Director of Policy, River Basin Center, University of Georgia, GA.
- 02/20-07/22 Postdoctoral Associate, WA Franke College of Forestry & Conservation,  
University of Montana, MT.
- 09/19-02/20 Postdoctoral Associate, Nicholas School of the Environment, Duke University,  
NC.
- 06/18-06/19 Instructor, School of Sustainability, Arizona State University, AZ.
- 08/16-05/18 Research Associate, School of Sustainability, Arizona State University, AZ.
- 01/15-05/16 Teaching Assistant, School of Sustainability, Arizona State University, AZ.

## Research Activities

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Order of names is as published and does not necessarily indicate priority of authorship. Papers where authorship is equally shared between myself and one or more authors are indicated by a \*. Publications involving students or postdocs are indicated with a †. Publications since the start of my tenure-track position are indicated by a ‡.

### Peer-Reviewed Journal Articles

20. Sarr, M., Mensah, E., **Vallury, S.**<sup>‡</sup>, Kapko, A., and Ojong, M. (2025). “Rethinking Resilience: Lessons from Climate-Smart Innovation in African Agriculture.” *Accepted at Journal of Development Perspectives*.
19. **Vallury, S.**<sup>‡</sup>, Nelson, D. R., and Cook, N. J. (2025). “Rethinking adaptation interventions in agricultural systems for sustainability.” *Current Opinion in Environmental Sustainability*, 77, 101571. <https://doi.org/10.1016/j.cosust.2025.101571>.
18. Benedum, M.E.<sup>†</sup>, Cook, N.J., and **Vallury, S.**<sup>‡</sup> (2025). “Out-Migration and Participation in Community-Based Natural Resource Management: The Role of Remittances.” *Ecology & Society*.30(3). <https://doi.org/10.5751/ES-16436-300334>.
17. **Vallury, S.**<sup>‡</sup>, Cook, N.J., and Nelson, D. (2024). “Social inequalities shape climate change adaptation in Indian farmers.” *Environmental Research Letters*. <https://doi.org/10.1088/1748-9326/ad7d1d>.

16. Olivier, T.\* and **Vallury, S.\*<sup>‡</sup>** (2024). “Institutional Fit and Policy Design in Water Governance: Nebraska’s Natural Resources Districts.” *Policy Studies Journal*. <https://doi.org/10.1111/psj.12550>.
15. Nesbitt, H., Metcalf, A., Floyd, T., Uden, D., Chaffin, B., Gulab, S., Banerjee, S., **Vallury, S.<sup>‡</sup>**, Hamlin, S., Metcalf, E., Fogarty, D., Twidwell, D., and Allen, C. (2024). “Social networks and transformative behaviors in a grassland social-ecological system.” *People & Nature*. <https://doi.org/10.1002/pan3.10695>.
14. Smith, A.P.<sup>†</sup>, **Vallury, S.<sup>‡</sup>**, and Metcalf, E.C. (2023). “Social dimensions of adaptation in rangeland social-ecological systems: A systematic literature review.” *Climatic Change*, 176, 180.
13. **Vallury, S.<sup>‡</sup>**, Chaffin, B. C., Hamlin, S. L., and Allen, C. R. (2023). “Communication in the science-policy interface: Evidence from a boundary organization in Nebraska, USA.” *Environmental Science & Policy*, 148, 103558.
12. Shin, H.C.<sup>†</sup>, Yousefi, P.<sup>†</sup>, Park, S.<sup>†</sup>, Yu, D.J., Janssen, M., **Vallury, S.<sup>‡</sup>**, and Araral, E. (2023). “Coping with Unreliable Water Supply: An Experimental Study of Exit and Voice.” *Water Resources Research*, 59, e2022WR032468.
11. Breetz, H.L., Stokes, L.C.\*, **Vallury, S.\***, and Cuiffo, K.V. (2022). “City Pledges for 100% Renewable Energy: A Matched Pair Analysis of Policy Adoption.” *Energy Research & Social Science*, 90, 102664.
10. Shin, H.C.<sup>†</sup>, **Vallury, S.**, Janssen, M.A., and Yu, D.J. (2022). “Joint effects of voluntary participation and group selection on the evolution of altruistic punishment.” *PLoS ONE*, 17(5), e0268019.
9. **Vallury, S.**, Smith, A.P., Chaffin, B.C., Nesbitt, H.K., Lohani, S., Gulab, S., Banerjee, S., Floyd, T.M., Metcalf, A.L., Metcalf, E.C., Twidwell, D., Uden, D.R., Williamson, M.A., and Allen, C.R. (2022). “Beyond the household: a literature review of scale and aggregation in adaptive capacity research”. *Environmental Research Letters*, 17, 063001.
8. **Vallury, S.**, H. C. Shin, M. A. Janssen, R. Meinzen-Dick, S. Kandikuppa, K. R. Rao, and R. Chaturvedi. (2022). “Assessing the institutional foundations of adaptive water governance in South India”. *Ecology and Society*, 27(1):18.
7. Shin, H.C., **Vallury, S.**, Abbott, J.K., Anderies, J.M., and Yu, D. (2022). “Understanding the role of competitive and participatory institutions in sustainable governance of irrigation systems”. *Ecological Economics*, 191, 107221.
6. **Vallury, S.** and Leonard, B.J. (2022). “Canals, Climate, and Corruption: The Provisioning of Public Infrastructure under Uncertainty”. *Economics & Politics*, 34, 221-252.
5. **Vallury, S.**, Abbott, J. K., Shin, H. C., and Anderies, J. M. (2020). “Sustaining Coupled Irrigation Infrastructures: Multiple Instruments for Multiple Dilemmas.” *Ecological Economics*, 178, 106793.
4. Mueller, V., Masias, I.\*, and **Vallury, S.\*** (2019). “Labor-Saving Technologies and Structural Transformation in Northern Ghana.” *Agricultural Economics*, 50(5), 581-594.
3. Janssen, M., Anderies, J.M., Baeza, A., Breetz, H.L., Jasinski, T., Shin, H.C., and **Vallury, S.** (2019). “Highways as coupled infrastructure systems: an integrated approach to address sustainability challenges.” *Sustainable and Resilient Infrastructure*, 1-12.
2. Baggio, J.A., Schoon, M., and **Valury, S.** (2019). “Managing networked landscapes: conservation in a fragmented, connected world.” *Regional Environmental Change*, 19(8), 2551-2562.
1. Bernstein, M.J., Mancha-Cisneros, M., Tyson, M., Brady, U.\*, Shin, H.C.\*, **Vallury, S.\***, Smith-Heisters, S., Ratajczyk, E. (2019). “Mapping Ostrom’s Common-pool Resource Systems Coding Handbook to the Coupled Infrastructure Systems Framework to Enable Comparative Research.” *International Journal of the Commons*, 13(1), 528-52.

## Manuscripts in Review

**Vallury, S.** and Kaur, R. “Spatial Inequalities in U.S. Electric Vehicle Infrastructure.” *Under review at Energy Policy*.

## Work in Preparation

Zhang, F., **Vallury, S.**, and Cook, N.J. “The Limits of Scaling Justice in U.S. Climate Adaptation Plans.”

**Vallury, S.**, Cook, N.J., Nelson, D., and Rao, Shishir. “Beyond Adaptive Capacity: Identity in Climate Migration.”

**Vallury, S.**, Cook, N.J., Paudel, S., and Koontz, T. “Can Information Motivate Conservation? Insights from a Behavioral Experiment.”

## Teaching Activities

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### Courses Taught

- Ecological Basis of Environmental Issues (Spring 2025)
- Agent-Based Modeling (Fall 2023, 2025)
- Environmental Policy & Natural Resource Management (Fall 2023, 2024, 2025)
- Water Policy Seminar (Spring 2023)
- Introduction to Applied Math for the Life & Social Sciences (Summer 2017-2019; Fall 2018)
- Economics of Sustainability (Spring 2019)
- Systems, Dynamics & Sustainability (Spring 2019)

## Research Grants

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|           | USDA NIFA Foundational Program. “Farmers’ Adoption Research and Choice Modeling of Agrivoltaics (FARM-AV).” <i>In Review</i> . (PI, \$300,000).                                |
| 2024-2027 | NSF Human-Environment and Geographical Sciences Program. “Equitable pathways to climate-smart agriculture: Rethinking adaptation as a continuum.” (PI, \$400,000).             |
| 2024-2025 | US Department of Interior. “2025 Biennial Georgia Water Resources Conference” (PI, \$25,000).                                                                                  |
| 2024-2027 | US Department of Army ERDC. “Expanding Research Test Beds for Nature-based Solutions in the Southeastern U.S.E US.” (Co-PI, \$5,484,643).                                      |
| 2024-2027 | NSF-Belmont Forum. “Mitigation and adaptation in cultural heritage landscapes: Lessons from transhuman pastoral systems for managing novel climate risks.” (Co-PI, \$399,906.) |
| 2023-2024 | NSF CRISES Program. “Planning Proposal: Collaborative Research and Education Center for Transformative Water Solutions.” (Co-PI, \$100,000).                                   |
| 2023-2024 | UGA E-Mobility Initiative Seed Proposal Program (Co-PI, \$65,000).                                                                                                             |
| 2023-2024 | UGA Teaming for Interdisciplinary Research Pre-Seed Grant (PI, \$4,000).                                                                                                       |
| 2018-2019 | Graduate Dissertation Fellowship, Center for Study of Economic Liberty, Arizona State University, AZ. (\$10,000).                                                              |

- 2018-2019 Humane Studies Fellowship, Institute for Humane Studies, (\$5,000).
- 2016-2017 Neely Foundation Food & Agriculture Sustainability Research Grant, School of Sustainability, Arizona State University, AZ. (\$4,000).

## Awards & Fellowships

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- 2024 The Institute for Humane Studies, Travel Fund (PI, \$750)
- 2023-2024 UGA Teaching Academy Early Career Fellows Program.
- 2021-2022 Hayek Travel Fund, Institute for Humane Studies, (\$1600).
- 2018-2019 Adam Smith Fellowship, Mercatus Center, (\$10,000).

## Conference Participation

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- “Access Inequalities in Electric Vehicle Infrastructure in the Southeast U.S.”. Midwest Political Science Association Annual Conference. Chicago, Illinois, USA. April 4-6, 2025.
- “Differentiated Effects of Climate Migration”. National Sustainability Society Symposium. Seattle, Washington, USA. September 9-11, 2024.
- “Social inequalities shape climate change adaptation among Indian farmers”. 2023 Association of Public Policy Analysis & Management. November 9-11, 2023.
- “Institutional Fit and Policy Design in Water Governance: Nebraska’s Natural Resources Districts”. 2023 Association of Public Policy Analysis & Management. November 9-11, 2023.
- “Social inequalities shape climate change adaptation among Indian farmers”. XIX Biennial IASC Conference, Nairobi. June 19-24, 2023.
- (*Invited*) Panel on “Natural Infrastructure and Equity”. 2023 Georgia Water Resources Conference. March 30-31, 2023.
- “Assessing the institutional foundations of adaptive water governance in South India”. 2022 Toronto Conference on Earth System Governance. October 20-24, 2022.
- “Beyond the household: a literature review of scale and aggregation in adaptive capacity research”. International Association for Society and Natural Resources Virtual Conference. 20-22 June, 2021.
- “Institutions Without Romance: Corruption and Provision of Shared Infrastructure in Canal Irrigation Systems.” 6th Workshop on the Ostrom Workshop, Indiana University, Bloomington, June 19-21, 2019.
- “Technology Adoption in the Presence of Shared Infrastructure: The Case of Long-Enduring Tank Irrigation Systems in India.” 2018 Utrecht Conference on Earth System Governance, November 5-8, 2018.
- (*Invited*) “Technology Adoption in the Presence of Shared Infrastructure: The Case of Long-Enduring Tank Irrigation Systems in India.” IHS Graduate Research Colloquium, Arlington, VA, October 12-14, 2018.
- “Institutions Without Romance: Corruption and Provision of Shared Infrastructure in Canal Irrigation Systems.” 55<sup>th</sup> Annual Meetings of the Public Choice Society, Charleston, SC, March 1-3, 2018.
- “Labor-Saving Technologies and Structural Transformation in Northern Ghana.” Workshop on Gender, Agricultural Growth, and Rural Transformation Research in the CGIAR, International Food Policy Research Institute, Washington D.C., December 14-15, 2017.

## Student Mentoring

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### *Current Graduate Students*

- Mackenzie Hallmark, Advisor, M.S. Ecology Integrative Conservation and Sustainability.
- Kaili Gregory, Committee Member, Ph.D., Warnell School of Forestry & Natural Resources.
- Monika Giri, Committee Member, Ph.D., ICON Anthropology.
- Maria Ramsey, Committee Member, Ph.D., Department of Agricultural Leadership, Education, & Communication.
- Michaela Collins, Committee Member, M.S. Warnell School of Forestry & Natural Resources.
- Niketa Nandan, Committee Member, M.S. Civil and Environmental Engineering.

### *Past Graduate Students*

- Jonathan Parrish, Advisor, M.S. Ecology Integrative Conservation and Sustainability. *Graduated Spring 2025.*
- Crystal Pendargast, Committee Member, M.S. Integrative Conservation and Sustainability. *Graduated Spring 2024.*
- Sabyasachi Kar, Committee Member, M.S. Warnell School of Forestry & Natural Resources. *Graduated Spring 2024.*
- Megan Seidel, Committee Member, MNR Warnell School of Forestry & Natural Resources. *Graduated Fall 2023.*

### *Undergraduate Students*

Samantha Dilley, 2023 Spring CURO Research Award

## Professional Service

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- Associate Editor, Ecology & Society
- Associate Editor, Journal of Development Perspectives
- Reviewer of Journals:
  - ◊ Policy Studies Journal
  - ◊ World Development
  - ◊ Ecological Economics
  - ◊ Environmental Science and Policy
  - ◊ International Journal of Commons
- Grant Reviewer, 2023 Social Sciences and Humanities Research Council of Canada

## Administration

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- Program Chair, 2025 Biennial Georgia Water Resources Conference.
- Organizer, 2024 Climate & Research Slam.
- Organizer, 2024 Confluence: UGA Water Science and Policy Poster Symposium.
- Co-chair, 2023 Biennial Georgia Water Resources Conference
- Organizer, 2023 Confluence: UGA Water Science and Policy Poster Symposium.

## University Service

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- 2024-2027 University Council
- 2024-2027 Undergraduate Committee

## Professional Memberships

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- American Economic Association
- Midwest Political Science Association
- International Association for Society and Natural Resources
- International Association for the Study of the Commons

## Related Programming Skills

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- Simulation Software: R, MATLAB, NetLogo, XPPAUT
- Statistical Software: STATA, R
- Other: QGIS,  $\text{\LaTeX}$